UML 2 - Renting School Facility

Activity Diagram and Sequence Diagram

Time	Tools	Submit
About 2 hours	PlantUML	PDF/PNG diagrams and .puml source

Goal

You will describe how a school facility rental system works. You will make two UML diagrams. The first diagram shows the steps in the process (Activity Diagram). The second diagram shows how the software parts communicate (Sequence Diagram).

Scenario

A school wants to earn money by renting its facilities after school hours. People and companies can use a website to rent a gym, classroom, parking lot, or sports field. The renter chooses a facility and a date. The school checks the request. If the request is approved and payment succeeds, the renter receives a confirmation.

People and software parts

Name	Meaning
Renter	A person or company that wants to rent a facility.
Website	The system used to submit and track a rental request.
School Staff	Checks the request and approves or rejects it.
Payment Service	Processes the payment.
Email Service	Sends confirmation or rejection messages.

Words used in this assignment

Word	Simple meaning
Request	Information sent by the renter.
Available	The facility is free on the requested date and time.
Approve	School Staff accepts the request.
Reject	School Staff does not accept the request.
Guard	A condition written in brackets, such as [available].
Message	Information sent from one participant to another.

System rules

- The renter selects a facility, date, time, and event type.
- The website checks if the facility is available.
- If the facility is not available, the website tells the renter. The renter may choose a different date or stop.
- If the facility is available, the renter enters contact information and submits the request.
- School Staff reviews the request. Staff may approve or reject it.
- If staff rejects the request, the website records the rejection and sends an email to the renter.
- If staff approves the request, the Payment Service charges the renter.
- If payment fails, the website records the failure and sends an email to the renter.
- If payment succeeds, the website saves the booking and sends a confirmation email.

Important Model only the process from the renter submitting a request through approval, rejection, or payment result. Do not model how the payment company works inside.

Task 1: Activity diagram

Create one activity diagram for the complete rental process. Use swimlanes. A swimlane shows who performs each action.

Your diagram must show:

- Renter, Website, and School Staff as swimlanes. You may add Payment Service and Email Service if helpful.
- A start point and at least three end points: rejected, payment failed, and booking confirmed.
- A decision for [available] and [not available].
- A decision for [approved] and [rejected].
- A decision for [payment succeeds] and [payment fails].
- The main actions in the correct order.
- Short action names, such as “Choose facility” or “Check availability”.

Task 2: Sequence diagram

Create one sequence diagram for a normal request. Start when the renter submits an available facility request. End when the booking is confirmed.

Your diagram must show:

- These participants: Renter, Website, School Staff, Payment Service, and Email Service.
- Messages for submitting the request, checking availability, reviewing the request, approving it, processing payment, saving the booking, and sending confirmation.
- An alt fragment for approval versus rejection.
- An alt fragment for payment success versus payment failure.

PlantUML starter code

Replace the comments with your own model. You should use PlantUML.

```
@startuml
title School Facility Rental - Activity

|Renter|
start
:Choose facility and date;

|Website|
:Check availability;
if (available?) then (yes)
  :Create request;
else (no)
  :Show not available;
  stop
endif

... add the rest ...
@enduml
```

```
@startuml
title School Facility Rental - Sequence

actor Renter
participant Website
participant "School Staff" as Staff
participant "Payment Service" as Pay
participant "Email Service" as Email

Renter -> Website: submit request
Website -> Staff: send request for review

alt approved
  Staff --> Website: approved
  ... add payment and confirmation ...
else rejected
  Staff --> Website: rejected
end
@enduml
```

What to submit

File	Requirement
Activity diagram	PDF or PNG image.
Sequence diagram	PDF or PNG image.
PlantUML source	One or two .puml files that can be rendered again.
Assumptions	A short note, up to 100 words. Explain any choice not stated in the rules.

Final check

- Can a reader see what happens when the facility is unavailable?
- Can a reader see what happens when staff rejects the request?
- Can a reader see what happens when payment fails?
- Do both diagrams use the same names and terms?
- Does the PlantUML source render without errors?

Submit: two diagrams, PlantUML source, and a short assumptions note